INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

11 April 2024 (am)

Subject CS1 – Actuarial Statistics
Core Principles

Paper A

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

If you encounter any issues during the examination please contact the Assessment Team on T. 0044 (0) 1865 268 873.

© Institute and Faculty of Actuaries

1 The following table shows possible realisations, x, of a random variable, X, the probabilities for these realisations and the conditional expectations and variances of a random variable, Y, conditionally on $X = x$.

Realisation x	10	20	40
$P[X = x]$	0.4	0.4	0.2
$E[Y\|X = x]$	2	8	10
$\mathrm{Var}(Y\|X = x)$	1	2	3

(i) Identify which **one** of the following options gives the correct values of the unconditional expectations of X and Y:

A $E[X] = 20$ and $E[Y] = 6$
B $E[X] = 20$ and $E[Y] = 4$
C $E[X] = 24$ and $E[Y] = 6$
D $E[X] = 24$ and $E[Y] = 4$.

[2]

(ii) Calculate the unconditional variance of Y. [3]

[Total 5]

2 The preparation time, in minutes, for coffees in a popular coffee shop has the following density:

$$f(y) = \frac{8}{m^4} y e^{-\frac{4y}{m}}, \qquad y \geq 0, m > 0$$

(i) Show that this density can be written in the form of the exponential family, also determining $b(\theta), a(\varphi)$ and $c(y, \varphi)$. [4]

Two models, A and B, are proposed for the parameter m, the mean preparation time, for different types of coffee.

Model A: $\frac{1}{m} = u_i \quad i = 1, 2, 3, 4$

Model B: $\frac{1}{m} = \begin{cases} u & i = 1 \\ u + v & i = 2, 3, 4 \end{cases}$

where $i = 1, 2, 3, 4$ correspond to different types of coffee.

(ii) Comment on the mean preparation times for different types of coffee, proposed by the two models. [2]

[Total 6]

3 X and Y are discrete random variables with joint distribution given in the table below.

	$Y=-3$	$Y=-2$	$Y=-1$	$Y=0$
$X=1$	$\frac{2}{5}$	0	$\frac{1}{10}$	0
$X=2$	$\frac{1}{5}$	$\frac{1}{10}$	0	$\frac{1}{5}$

(i) Identify which **one** of the following options gives the correct value of the conditional expectation $E[Y|X=2]$:

A -2.2
B -1.8
C -1.6
D -0.3.

[2]

(ii) Determine the conditional expectation $E[X|Y=y]$ for each value of y. [4]

(iii) Calculate the expected value of X using your result from part (ii). [2]

[Total 8]

4 Consider random variables X, Y and Z, each of which has variance equal to 1.

Suppose that $\text{Cov}(X,Y)=0.9$ and $\text{Cov}(Y,Z)=0.8$.

Let $A=X-0.9Y$ and $B=Z-0.8Y$.

(i) Calculate:

(a) $\text{Var}(A)$

(b) $\text{Var}(B)$.

[3]

(ii) Show that $\text{Cov}(A,B)=\text{Cov}(X,Z)-0.72$. [3]

(iii) Show that $\text{Cov}(X,Z)\geq 0.458$. [3]

[**Hint:** use the fact that $\text{cor}(A,B)\geq -1$, where $\text{cor}(A,B)$ is the correlation coefficient between A and B.]

[Total 9]

5 Let $\underline{x} = (x_1, x_2, \ldots, x_n)$ be independent observations from a distribution of a random variable, X, that has parameter u. Let $f(u|\underline{x})$ be the posterior distribution of $u|\underline{x}$ and $g(\underline{x})$ be an estimator of u.

(i) State two properties that a loss function should satisfy. [1]

Consider the loss function $L(g,u)$ given by:

$$L(g,u) = \begin{cases} (g-u), & u \leq g \\ 3(u-g), & u > g \end{cases}$$

(ii) Verify that this loss function satisfies the properties in part (i). [2]

(iii) Identify which **one** of the following options gives the correct expression for the expected posterior loss:

A $\quad 4\int_{-\infty}^{g}(g-u)f(u|\underline{x})\,du + \int_{g}^{\infty}(u-g)f(u|\underline{x})\,du$

B $\quad \int_{-\infty}^{g}(g-u)f(u|\underline{x})\,du + 4\int_{g}^{\infty}(u-g)f(u|\underline{x})\,du$

C $\quad 3\int_{-\infty}^{g}(g-u)f(u|\underline{x})\,du + \int_{g}^{\infty}(u-g)f(u|\underline{x})\,du$

D $\quad \int_{-\infty}^{g}(g-u)f(u|\underline{x})\,du + 3\int_{g}^{\infty}(u-g)f(u|\underline{x})\,du.$

[3]

(iv) Identify which **one** of the following options gives the correct quantile for the Bayesian estimator, b, of u under this loss function:

A $\quad P(u < b|\underline{X}) = 0.2$

B $\quad P(u < b|\underline{X}) = 0.25$

C $\quad P(u < b|\underline{X}) = 0.75$

D $\quad P(u < b|\underline{X}) = 0.8.$

[**Note:** you may assume that, for any differentiable function $h(x,y)$ such that $h(x,x) = 0$:

$$\frac{d}{dx}\int_{x}^{\infty} h(x,y)\,dy = \int_{x}^{\infty}\frac{\partial}{\partial x}h(x,y)\,dy$$

and:

$$\frac{d}{dx}\int_{-\infty}^{x} h(x,y)\,dy = \int_{-\infty}^{x}\frac{\partial}{\partial x}h(x,y)\,dy]$$

[3]
[Total 9]

6 A lecturer wants to investigate the proportion, p, of students in a large class that are in favour of online lectures. The lecturer conducts two small surveys, where students can vote in favour or against online lectures, and the results are as follows:

Survey	*Sample size*	*In favour*
1	10	4
2	20	9

The two samples are assumed to be independent.

The lecturer does not initially have any prior information about the proportion, p, and assumes a suitable uniform distribution as the prior distribution in order to estimate p.

(i) State the parameters of a suitable prior distribution for p, justifying your answer. [2]

(ii) Determine the Bayes estimate of p under the quadratic loss function. [5]

The lecturer later decides instead to use a beta distribution as the prior, with parameters a and b, where $a = 2b$.

(iii) Determine the new Bayes estimate of p under the quadratic loss function, in terms of parameter b. [4]

The lecturer is interested in obtaining a posterior estimate involving low uncertainty.

(iv) Determine whether the estimate in part (ii) or part (iii) produces a posterior estimate with lower uncertainty, in the case where $b = 1$. [2]

[Total 13]

7 (i) Explain what is meant by a conjugate prior distribution. [1]

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables from the Poisson distribution with parameter m, and m follows, a priori, a gamma distribution with probability density function given by $f(m) = \frac{s^a}{\Gamma(a)} m^{a-1} e^{-sm}$, with $a, s > 0$.

(ii) Show that this prior distribution is conjugate for m. [3]

(iii) Determine the mean and variance of the posterior distribution of m. [2]

(iv) Comment on how the prior distribution affects the posterior for large sample sizes, n. [2]

Consider now the parameter $1/m$.

(v) (a) Identify which **one** of the following options gives the **prior** mean for parameter $1/m$:

A $E\left(\frac{1}{m}\right) = \frac{s-1}{a}$

B $E\left(\frac{1}{m}\right) = \frac{a-1}{s}$

C $E\left(\frac{1}{m}\right) = \frac{a}{s-1}$

D $E\left(\frac{1}{m}\right) = \frac{s}{a-1}$.

[3]

(b) Identify which **one** of the following options gives the **posterior** mean for parameter $1/m$:

A $E\left(\frac{1}{m} \mid x\right) = \frac{s+n}{a+n\bar{x}-1}$

B $E\left(\frac{1}{m} \mid x\right) = \frac{a+n\bar{x}-1}{s+n}$

C $E\left(\frac{1}{m} \mid x\right) = \frac{s+n}{a+n\bar{x}}$

D $E\left(\frac{1}{m} \mid x\right) = \frac{a+n\bar{x}}{s+n}$.

[3]

[Total 14]

8 In a particular year, the members of a regional oil organisation decide to increase the production of oil. An analyst wishes to model the effect of this increase on the market price of oil and recorded the monthly average volume produced, x, in litres (l), and its price per litre, y (£) for eight consecutive months. The data are summarised as follows:

$$S_{xx} = 3{,}535{,}237.5 \quad \sum y = 181 \quad \sum y^2 = 4305 \quad S_{xy} = -23{,}726.25$$

(i) Calculate Pearson's correlation coefficient between x and y. [3]

(ii) Comment on your answer to part (i). [1]

A second analyst thought that converting the volume to cubic metres (m^3), as $z = \frac{x}{1000}$, would make the computation more efficient.

(iii) Write down S_{zz} expressed in terms of S_{xx}. [2]

(iv) Calculate the new Pearson's correlation coefficient between z and y. [2]

(v) Comment on your answers to parts (i) and (iv). [1]

(vi) Determine the least squares fitted regression line between y and z given that $\sum z = 0.16$. [3]

(vii) Determine a 95% prediction interval for the price of oil, when the oil production is 1.5 m^3, based on the observed data. [4]

[Total 16]

9 A small vehicle repair shop, Repair Shop 1, has recently merged with another nearby vehicle repair shop, Repair Shop 2, and the owner wishes to analyse the number of customers that visit each shop.

The owner has recorded the following data over the past 30 days:

	Number of customers	
Day	*Repair Shop 1*	*Repair Shop 2*
1	30	22
2	31	24
…		
30	29	26
Total	**781**	**707**

(i) Calculate the sample means of the number of daily customers for each repair shop. [1]

The owner believes that the number of customers arriving at Repair Shop 1 and Repair Shop 2 per day follow separate Poisson distributions with unknown parameters k_1 and k_2, respectively.

(ii) Calculate a 95% confidence interval for the difference between the parameters k_1 and k_2, stating any assumptions you make. [5]

(iii) Test at the 5% significance level whether k_1 and k_2 are equal. [2]

The owner later believes it may instead be more appropriate to consider the recorded data as being paired data.

(iv) Explain why it may be more appropriate to treat the data as paired. [3]

The sample standard deviation of the paired differences has been recorded as 6.55.

(v) Test at the 5% significance level whether k_1 and k_2 are equal, assuming the data is paired, stating any additional assumptions you make. [5]

(vi) Comment on your answers to parts (iii) and (v). [4]

[Total 20]

END OF PAPER